

Latency in Distributed LLM Inference

PROGRESS UPDATE: JUNE 24, 2026

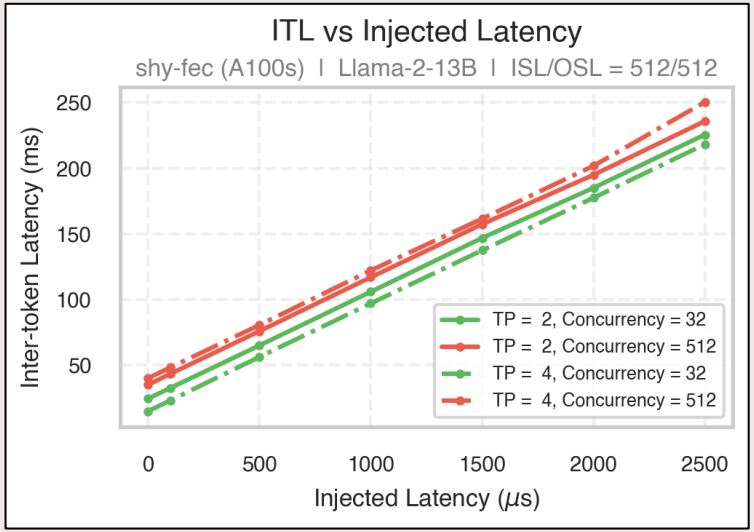
Kelsey Penners

Status / Progress

- *Power Analysis:* processed power logs from all runs, computed joules per token with `nvidia-smi` data sampled at 20 Hz
- *Latency Injection:* using `torch.cuda._sleep()` to make GPU idle for N clock cycles, converted from target µs to inject using device clock rate, swept 0 – 2.5ms at concurrency 32/512 for 8b/13B, TP=2/4
- *SLOs:* computed max injected latency or max concurrency before system no longer attains SLO, can easily compute for any config

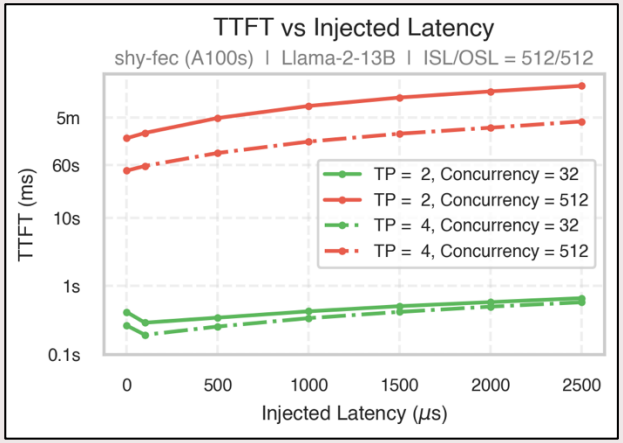
Latency Injection in Inter-Token Latency (ITL)

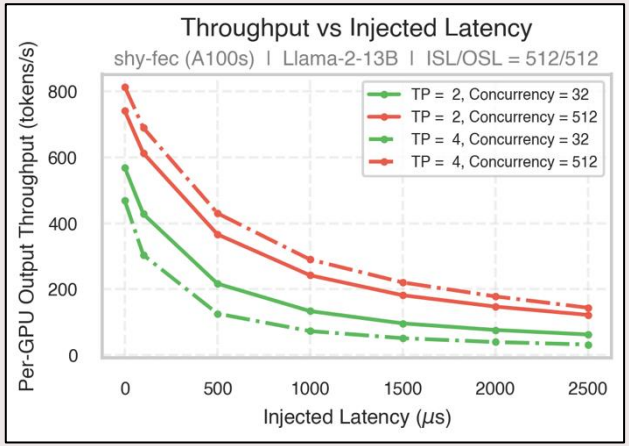
- Expected ITL degradation scales linearly with AllReduce injection:
  $\Delta\, ITL = 2 \times \text{layers} \times \Delta\, \text{AR latency}$
- Should see slope of 80 ms/ms for 13B (40 layers) and 64 ms/ms for 8B (32 layers)
- Observed data stays within 5% error, other than at very small injections
- Validates that injection is working as expected!



$\text{Injected latency} = \Delta\, \text{AR latency}$

Decode vs Prefill Under Latency Injection

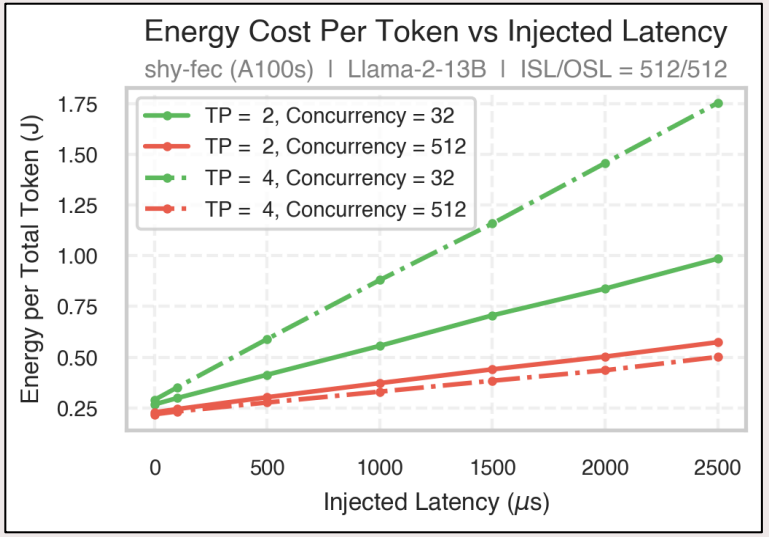




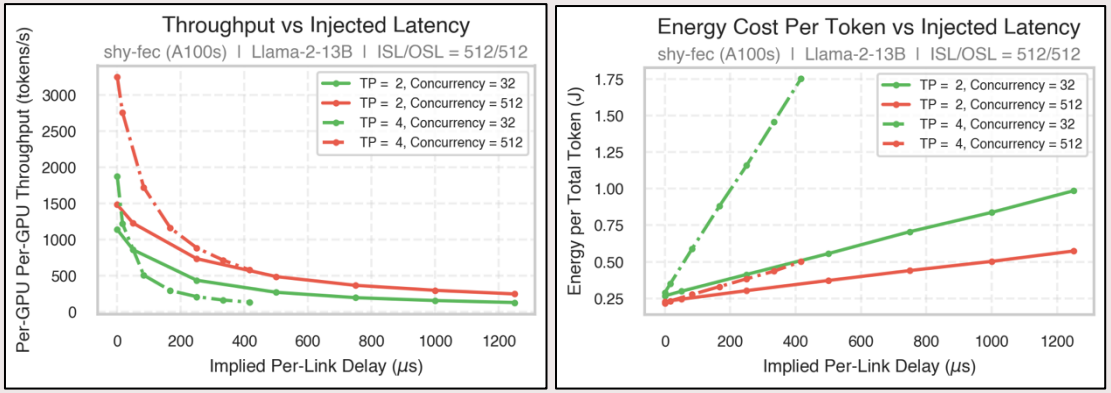
- At low concurrency, TTFT stays relatively flat across injection
- After KV cache saturation, request queueing causes TTFT increase
- For throughput, TP=4 is most sensitive to added latency

Energy Cost vs Injected Latency

- Energy per token scales linearly with injected latency for all configs
- At 2.5 ms added,
  - Low concurrency: TP=4 largest increase (500%), TP=2 (275%)
  - High concurrency: both configs (120-150%)
- Higher concurrencies are more energy efficient
- Slope reveals sensitivity to injected latency; TP 4, concurrency 32 most sensitive



Issue with Extrapolating to Larger Systems



- Dividing injected delay by hop count (2(N-1)) to extrapolate doesn't work well
- Scaling hardware changes compute power, not just communication characteristics
- Claude code simply transformed x-axis, doesn't provide accurate predictions

Discussion

- Implied per-link added latency to extrapolate to larger GPU systems
- SLOs: high concurrency configs fail to meet SLOs due to system saturation
- TP 8?

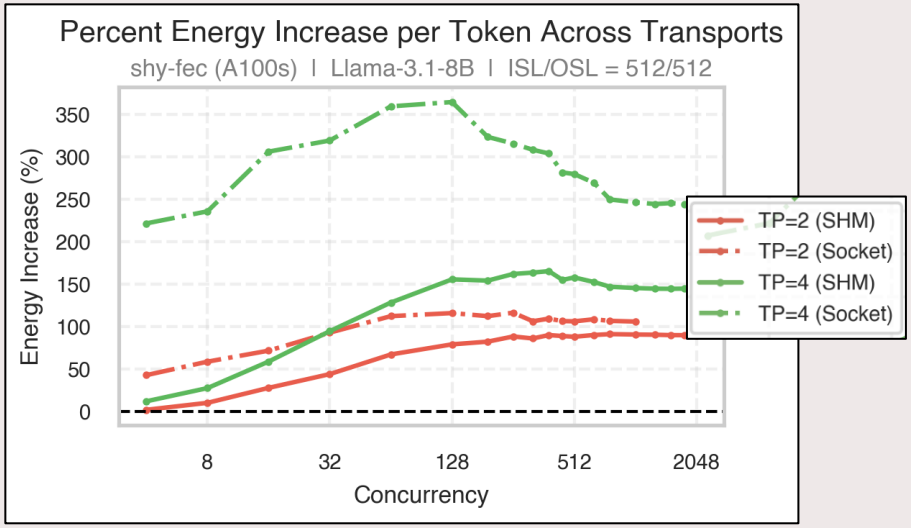
Model	TP	Concurrency	Max injected latency before SLO violation
Llama 3.1 8B	2	32	1005 µs
	4		720 µs
Llama 2 13B	2		127 µs
	4		714 µs

Next Steps

- Holiday ☺
- Writing results for multi-GPU chapter
  - Will cover different transports & latency injection
- Creating diagrams for report

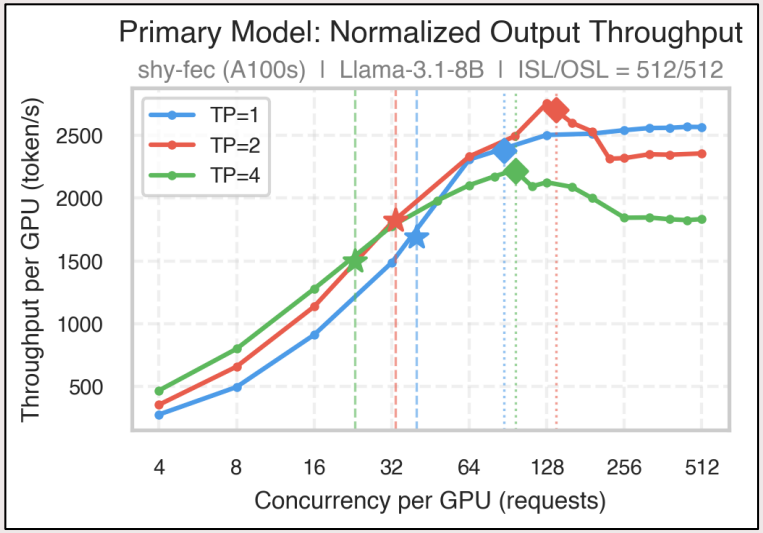
Thank you!

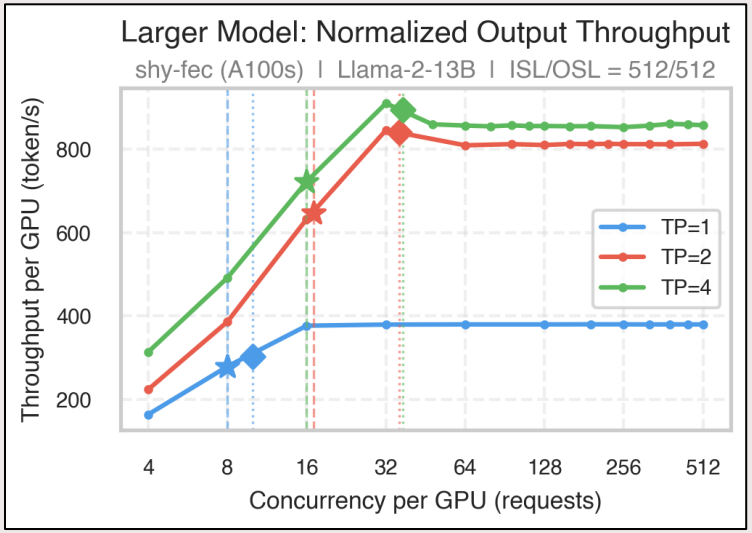
Different Transports, Different Energy Consumption



- TP4 suffers larger energy penalty compared to TP2
- For TP2, SHM and socket transport have similar energy profiles
- Scaling to TP4 increases gap between SHM and socket
  - More AllReduce hops

Representing SLOs on Plots





☆ = reaches responsive SLOs
◇ = reaches acceptable SLOs

P95 SLOs:

Metric	Acceptable SLO (ok)	Responsive SLO (good)
TTFT	5 s	1 s
ITL	0.25 s	0.1 s

TU/e